Distributed Data Processing (DDP) refers to : The approach of handling and analyzing data across multiple interconnected devices.

Major forces contributed in the development of the DDP

1. The organizational structure of large corporations that becomes fragmented through geographical distribution.
2. The focus on information as valuable resource.
3. The technological potential and cost effectiveness of handling information in a new way.

2. Define data processing systems? .... "Choose any one you prefer "

Broad definitions and emphasize the relationship between goals and resources:

- The concept of arranging hardware, software, and networks to meet organizational needs.
- The philosophy of placing computers geographically and in organization closes to the application.
- Putting resources where the people are.

Emphasize the processing needs of distributed systems:

- The use of multiple computers in a cooperative arrangement to support common systems objectives.
- The transfer of data processing functions from a central system to the small-dedicated processors where the information is used.

Focus on the software and data aspects of DDP:

- Programs and data reside in processing nodes that are integrated by networks. The interconnection of components of a system permitting shared resources and databases.

Very specific - defines DDP in 3 words: distributed, data, and processing:

- Placement of intelligent devices at multiple sites to execute user-written applications to update and store local files and accesses remote databases.

Very specific - identify functional responsibility:

- A local site connected to a remote host via a network in order to access corporate database and provide processing, data entry, and printing.
- Combines both local and remote processing and reflect the more recent preoccupation with productivity tools for end user:
- The provision of an integrated computer network for end-user productivity.

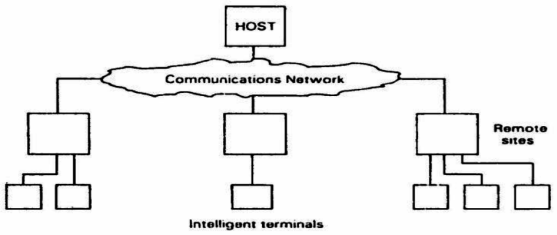
Benefits of DDP:

1- Scalability
2- Fault tolerance
3- Performance
4- Efficient handling of large volumes of data.

4. **Explain the difference between data processing system (Architecture) – [Vertical - Horizontal and Functional]?**

Vertical Topology:

There are three levels of processing at the host, at the remote site, and in intelligent terminals.

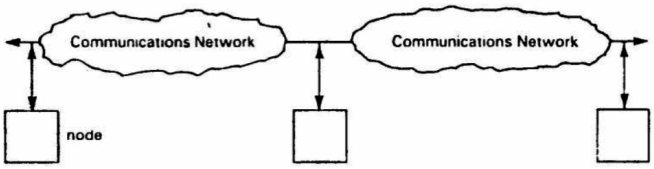


1. Transactions enter and leave at the lowest levels.
2. Processing not possible at the intelligent terminal is passed to the remote-site processor.
3. Processing not possible at the remote site is passed to the host or primary site.
4. Each level may support its own database, which may or may not be shared with other levels.

Horizontal Topology:

Horizontal topology is characterized by peer (equal) relationships among nodes.

Every node in the network can communicate with every other node without having to consult a central node.

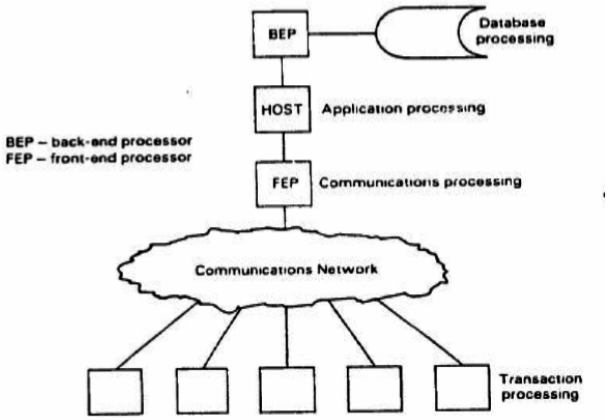


Functional Topology:

This topology involves the separation of processing functions into separate nodes.

Typical functional separations are:

1. Database processing,
2. Backup processing,
3. Application processing,
4. Communication processing, and transaction processing.



5. **What are the Advantages and Disadvantages of DDP ?**

Advantages :

1- Short response time.
2- End user application control.
3- Flexibility.
4- Less cost of medium data processing.

Disadvantages :

1- Maintenance cost.
2- Complexity.

6. Mention the main problem that faces the development of distributed data processing systems?

The main problem that faces the development of distributed data processing systems:

1) Data Recovery : Solution – Transaction management must track transaction to ensure the correct termination of the transaction.

2) Concurrency Control : Solution – Control access of several transactions on the same data

3) Deadlock /Lockup Handling : Solution –To prevent transactions from waiting each other's